

Python File Handling

Back-to-Basics Series

Python File Handling

Text File

- Open, Read
- Create, Write, Append

CSV File

- Open, Read
- Write

Part 1 : Reading a Text File

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Open a Text File in Read mode

Method 1:

```
f = open("textfile.txt")    #read mode by default  
f = open("textfile.txt", "r")  
f.close()
```

Method 2:

```
with open("textfile.txt") as f:
```

Open a Text File with Exception

Method 1:

```
try:
    f = open('textfile.txt')
except Exception as e:
    print(e)
else:
    data = f.read()
    print(data)
    f.close()
```

If the file does not exist, it will show an error message:
[Errno 2] No such file or directory: 'textfile.txt'

Must always close the file.

Open a Text File with Exception

Method 2:

try:

```
with open('textfile.txt') as f:  
    data = f.read()  
    print(data)  
    #f.close()
```

No need to
close the file.

```
except Exception as e:  
    print(e)
```

If the file does not exist, it will
show an error message:
**[Errno 2] No such file or directory:
'textfile.txt'**

3 Ways of reading a Text File

`read(<size>)` : read into a string ('size' is optional)

`readline()` : read only the first line

`readlines()` : read all the lines into a list.

* using `.strip()` to remove the newline `'\n'`.

Read a Text File

```
with open("textfile.txt", "r") as f:  
    content = f.read()  
    print(content)
```

- "r" : read
- "type (content)" : String

Read a Text File – line by line

```
with open("textfile.txt", "r") as f:  
    content = f.readlines()  
    for line in content:  
        print(line)
```

- "r" : read
- "type(content)" : List
- "line" : contains new line character "\n"

Read a Text File – remove "\n"

```
with open("textfile.txt", "r") as f:  
    content = f.readlines()  
    for line in content:  
        print(line.rstrip('\n'))
```

- "r" : read
- "type(content)" : List
- "rstrip" : remove "\n"

Close a Text File

```
f.close()
```

- closing the file will immediately free up any system resources used by it.

Part 2 : Creating, Writing or Appending to a Text File

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Create/Overwrite a Text file

```
f = open("writefile.txt", "w")  
f.close()
```

- a new file will be created if it does not exist.
- if a file with the same filename exists, then it will be overwritten.

Writing into a Text File

```
f = open("writefile.txt", "w") :  
  
for i in range(10):  
    f.write("This is line", i)  
  
f.close()
```

```
This is line 0  
This is line 1  
This is line 2  
This is line 3  
This is line 4  
This is line 5  
This is line 6  
This is line 7  
This is line 8  
This is line 9
```

Append to an existing Text File

```
f = open("writefile.txt", "a") :  
  
for i in range(10):  
    f.write("This is line", i+10)  
  
f.close()
```

```
This is line 0  
This is line 1  
This is line 2  
This is line 3  
This is line 4  
This is line 5  
This is line 6  
This is line 7  
This is line 8  
This is line 9  
This is line 10  
This is line 11  
This is line 12  
This is line 13  
This is line 14  
This is line 15  
This is line 16  
This is line 17  
This is line 18  
This is line 19
```

Part 3 : Reading a CSV File

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Common Separated Values (CSV) File

The CSV (Comma Separated Values) format is the most common import and export format for spreadsheets and databases. It is a plain text file that contains a list of data separated by commas.

The content of a CSV file may be read using any text editor, eg the Notepad.

Open a CSV File

Method 1:

```
f = open("datafile.csv")  
f.close()
```

Method 2:

```
with open(" datafile.csv") as f:
```

Read a CSV File with header

```
from csv import *  
  
f = open("datafile.csv")  
content = reader(f)  
for row in content:  
    print(row)
```

– "type(row)" : List

Read a CSV File without header

```
from csv import *  
  
f = open("datafile.csv")  
content = reader(f)  
next(content)  
for row in content:  
    print(row)
```

– "next () " : skip the first row

Read a CSV File – individual fields

```
from csv import *  
  
f = open("datafile.csv")  
content = reader(f)  
next(content)  
for name, gender, ht, wt in content:  
    print(name, gender, ht, wt)
```

Part 4 : Writing to a CSV File

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Create/Overwrite a CSV file

```
f = open("file_does_not_exist.csv", "w")  
f.close()
```

- a new file will be created if it does not exist.
- if a file with the same filename exists, then it will be overwritten.

Write to a CSV File

```
fields = ['Name', 'Class', 'Level', 'Score']
```

```
data = [['Nick', 'S15', '15', '5460'],  
        ['Mary', 'S16', '12', '6105'],  
        ['Peter', 'S15', '10', '3700']]
```


Write to a CSV File

```
from csv import *
```

```
f = open("records.csv", "w", newline = '')
```

```
content = writer(f)
```

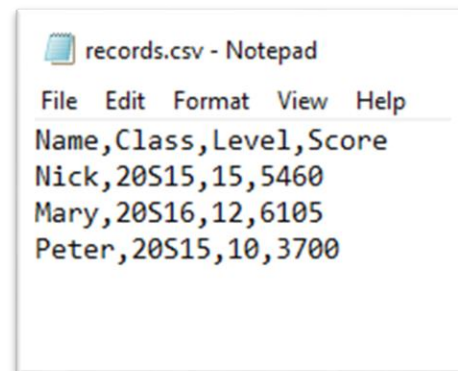
```
content.writerow(fields)
```

```
content.writerows(data)
```

```
f.close()
```

	A	B	C	D
1	Name	Class	Level	Score
2	Nick	20S15	15	5460
3	Mary	20S16	12	6105
4	Peter	20S15	10	3700

To remove the extra *newline* introduced.



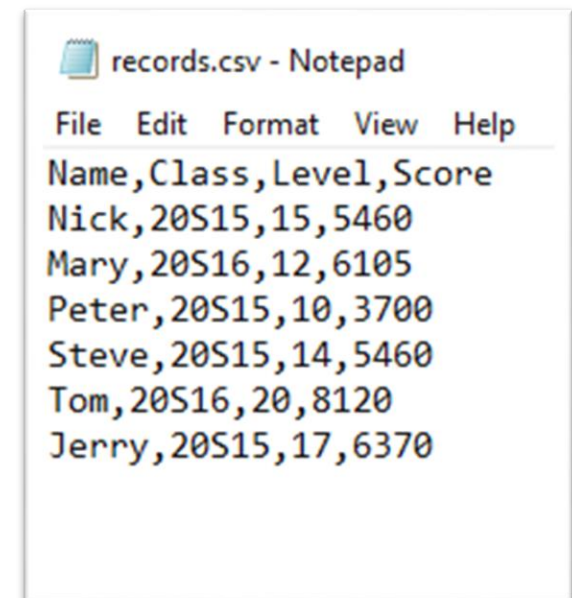
Append to a CSV File

```
more_data = [['Steven', 'S15', '14', '5460'],  
             ['Tom', 'S16', '20', '8120'],  
             ['Jerry', 'S15', '17', '6370']]
```

```
from csv import *
```

```
f = open("records.csv", "a") :  
content = writer(f)  
content.writerows(more_data)
```

```
f.close()
```



The End
